

Using the Set Operators

1. Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job, for departments 10, 20, and 30, giving each column an appropriate heading.

Solution:-

```
SELECT job,  
       SUM(DECODE(department_id,10,salary)) AS Dept10,  
       SUM(DECODE(department_id,20,salary)) AS Dept20,  
       SUM(DECODE(department_id,30,salary)) AS Dept30,  
       SUM(salary) AS Total  
FROM employees  
WHERE department_id IN (10,20,30)  
GROUP BY job;
```

2. Using the SET operator, display the DEPTNO, SUM(SAL) for each department, JOB, SUM(SAL) for each job, and Total Salary.

Solution:-

```
SELECT TO_CHAR(department_id) AS Department,  
       NULL AS Job,  
       SUM(salary) AS Total_Salary  
FROM employees  
GROUP BY department_id
```

UNION

```
SELECT NULL,  
       job_id,  
       SUM(salary)  
FROM employees  
GROUP BY job_id
```

UNION

```
SELECT 'TOTAL',  
       NULL,  
       SUM(salary)  
FROM employees;
```

3. Using the SET operator, display the JOB and DEPTNO of employees working in departments 20, 10, and 30 in that order.

Solution:-

```
SELECT job_id, department_id  
FROM employees  
WHERE department_id = 20
```

UNION ALL

```
SELECT job_id, department_id  
FROM employees  
WHERE department_id = 10
```

UNION ALL

```
SELECT job_id, department_id  
FROM employees  
WHERE department_id = 30;
```